

Anjali S Rameja, PhD

Computational Physicist | Physics-Aware Machine Learning for Atomistic Simulation

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SCIENTIFIC MACHINE LEARNING

- Atomistic foundation models: MACE-MPA-0 fine-tuning
- ML interatomic potentials as surrogates for DFT
- Training-set design, DFT labelling, active-learning reserve
- Validation against classical potentials and experiment
- Throughput, scaling, and precision benchmarking

PHYSICS-BASED SIMULATION

- Molecular dynamics (LAMMPS): 2NN-MEAM, EAM, Stillinger-Weber, tight binding; 10,000-atom systems, 300 to 5800 K
- Density functional theory (Quantum ESPRESSO, USPP, convergence protocols)
- ASE for structure building and workflows

HPC AND DEPLOYMENT

- SLURM, multi-node A100 GPU cluster (PARAM Rudra, NSM); Nsight Systems profiling
- GPU builds: LAMMPS + libtorch + Kokkos + CUDA 12.4
- NVIDIA NGC containers, MPI (OpenMPI, srun pmix)
- Linux, bash, job orchestration for hundreds of runs

PROGRAMMING

- Python (NumPy, SciPy, Pandas, ASE)
- PyTorch (MACE training and inference stack)
- Git and version control

AI TRAINING AND EVALUATION

- Turing, Aligned, Handshake AI (2026)
- Scientific coding, agentic terminal, and multimodal physics tasks; rubrics and hidden test suites

PROFESSIONAL SUMMARY

Computational physicist (PhD) building physics-aware machine-learning surrogates for atomistic simulation. Six years of molecular dynamics and DFT on liquid and glassy metals, six peer-reviewed papers (one Journal of Applied Physics Featured Article), and since 2026 an end-to-end scientific AI workflow: fine-tuning the MACE-MPA-0 atomistic foundation model on a purpose-built DFT dataset, deploying it in LAMMPS on multi-node A100 GPUs, and validating it against classical potentials and experiment. Owns the whole loop: data, training, validation, deployment, and physics judgement.

SCIENTIFIC AI PROJECT

Fine-tuning an Atomistic Foundation Model for Mg-Ca Metallic Glasses

Prof. N. K. Bhatt, MK Bhavnagar University | Jun 2026 to Present, in progress

- **Training-set design:** 2,000-geometry DFT set across four Mg-Ca compositions and crystal, liquid, supercooled and glassy states, labelled in two rounds so round two targets the model's errors.
- **DFT convergence protocol:** Quantum ESPRESSO settings converged on 18 cells with a worst-case criterion (40 Ry, dual 4, k-mesh rule by cell type), 270 calculations; reading crystals and disordered cells separately saved about 1,100 CPU node-hours.
- **Workflow engineering:** automated inputs, SLURM batching, provenance and collection; an in-job scheduler packs small calculations into the 10-node queue minimum, after 63% of billed time sat idle.
- **GPU deployment and performance:** LAMMPS with MACE (libtorch, Kokkos, CUDA 12.4) on A100 nodes, validated against ASE to numerical precision; 132,137 atom-steps/s per node at 96.4% weak-scaling efficiency on 8 nodes; one allocator setting doubled atoms per card and gave up to 3.3x throughput.
- **Profiling and debugging:** Nsight Systems shows a long-tailed, data-movement-bound profile; traced a multi-GPU segfault in the LAMMPS Kokkos ML-IAP path to host code dereferencing device memory, confirmed in NVIDIA's container, documented for upstream.
- **Physics validation:** zero-shot MACE reproduces the Mg-to-Ca icosahedral-centre ratio in the glass (3.3 against 5.6 from experiment) where the classical 2NN-MEAM potential gives 1.0.
- **Compute grant and reporting:** PARAM Rudra (NSM) A100 allocation, Co-Investigator, 2026; million-atom run priced at 1,261 GPU node-hours from measured throughput; first-month report delivered to C-DAC.

EXPERIENCE

Research Quality Specialist / Subject Matter Expert, Physics and Materials

Turing | Remote | June 2026 to Present (3 projects)

- Author and review paper-derived scientific coding tasks for frontier-lab clients across electrochemistry, Poisson-Boltzmann electrostatics, thermochemical equilibrium, rovibrational spectroscopy, and solid-state chemistry: reference solutions, rubrics, mutation-tested hidden test suites; agentic LAMMPS tool-use tasks scored on input decks and RDF, MSD, and convergence checks.

MODELING AND ANALYSIS

- Statistical mechanics and mathematical modeling
- Liquid-state theory
- Green-Kubo and Stokes-Einstein transport (diffusion, viscosity)
- Structure and phase transitions (glass transition, RDF, CNA, Voronoi)
- Time correlation functions
- Thermophysical and mechanical properties

EDUCATION

PhD, Physics (Computational Materials Science)

MK Bhavnagar University, Gujarat | 2020 to 2026

MSc, Physics (Condensed Matter)

Sardar Patel University, Gujarat | 2013 to 2015

BSc, Physics (ranked 3rd in cohort)

Sir P. P. Institute of Science, Gujarat | 2010 to 2013

ACHIEVEMENTS

PARAM Rudra GPU allocation (NSM), Co-Investigator | 2026

1st Prize: Oral Presentation

National Conference at IITE, Gandhinagar | 2022

2nd Prize: Poster Presentation

National Science Symposium, Govt. Sci. College, Gariyadhar | 2024

HIGHLIGHTS

- Young Researchers' Workshop on Machine Learning for Materials | ICTP | 2022
- 16th International Conference on Creep and Fracture of Engineering Materials and Structures | IISc | 2024
- Molecular Dynamics Simulations in Materials Science and Engineering | IITkgp | 2021
- Computational Physics and Materials Science: Total Energy and Force Methods | ICTP | 2021
- Introduction to Glass Science and Technology | IITkgp | 2020
- 9 conference presentations (2020 to 2024)

AI Trainer, Scientific Computing

Aligned | Remote | September 2026 to Present

- Build containerised terminal-bench style tasks (reference solution, hidden verifier) that test data triage and method choice, e.g. sound speed of a binary liquid from six LAMMPS NPT runs of unequal quality.

AI Trainer, Physics

Handshake AI | Remote | September 2026 to Present

- Author adversarial multimodal physics tasks (lab image plus expert prompt) that current vision-language models fail for a nameable physical reason.

Doctoral Researcher, Computational Materials Science

Department of Physics, MK Bhavnagar University | Supervisor: Prof. N. K. Bhatt | Feb 2020 to Jan 2026

- **Physics-based simulation at scale:** designed and ran large-scale LAMMPS molecular dynamics on Mg_2Ca , Al_2Ca , Al_4Ca , and $\text{Ca}_{61}\text{Al}_{39}$ (10,000 atoms, 300 to 1500 K, 2NN-MEAM) and on liquid W and Mo from melting to boiling point (2900 to 5800 K).
- **Property extraction:** diffusion coefficients, viscosity, glass-transition temperatures, full elastic suite, thermodynamic properties, and structural descriptors (RDF, $S(q)$, CNA) across glassy, supercooled, and liquid regimes; demonstrated Stokes-Einstein breakdown in supercooled metallic glasses.
- **Multi-fidelity cross-validation:** Quantum ESPRESSO calculations on the crystalline phases (electronic structure, bonding) used to check MD-derived thermophysical properties against ab-initio reference.
- **Analysis pipelines:** reusable Python (NumPy, SciPy, Pandas) batch post-processing of multi-temperature trajectories: RDF, MSD, VACF, $S(q)$, time-correlation functions ([public repo: metallic-glass-md](#)).
- **Application impact:** established Mg-Ca metallic glasses as candidate biodegradable bone-grafting implants (JAP Featured Article).

Visiting Faculty and Lecturer, Physics

MK Bhavnagar University and affiliated colleges | 2015 to 2024

- Taught computational and condensed matter physics (MSc), supervised LAMMPS-based MD dissertations, ran MD and visualization workshops.

PUBLICATIONS

1. A. Shankar, N. K. Bhatt. "[Atomic mass transport in binary metallic glasses: a case study of \$\text{Mg}_2\text{Ca}\$, \$\text{Al}_2\text{Ca}\$ and \$\text{Al}_4\text{Ca}\$](#) ." Journal of Physics: Condensed Matter 37, 355401 (2025).
2. A. Shankar, N. K. Bhatt. "[Enhanced mechanical and thermophysical properties of \$\text{Mg}_2\text{Ca}\$, \$\text{Al}_2\text{Ca}\$, and \$\text{Al}_4\text{Ca}\$ bulk metallic glasses in comparison to crystalline alloys for bone grafting applications](#)." Journal of Applied Physics 136(3) (2024). **[Featured Article]**
3. A. Shankar, N. K. Bhatt. "[Bulk metallic glass \$\text{Al}_2\text{Ca}\$ from a biomedical application viewpoint: A molecular dynamics study](#)." IOP Conference Series: Materials Science and Engineering 1291, 012026 (2023).
4. D. R. Gohil, A. Shankar, N. K. Bhatt. "[Mass transport and thermal properties of liquid \(melting to boiling point\) tungsten: a molecular dynamics simulation](#)." Physica Scripta 98, 115963 (2023).
5. D. R. Gohil, A. Shankar, N. K. Bhatt. "[Molecular Dynamics Simulations for Liquid Molybdenum: Structural and Atomic Transport Properties](#)." Bulgarian Journal of Physics 50(4) (2023).
6. A. Shankar, N. K. Bhatt, A. B. Patel, P. R. Vyas. "[Amorphous structure of binary \$\text{Ca}_{61}\text{Al}_{39}\$ metallic glass close to transition temperature](#)." AIP Conference Proceedings 2352, 020038 (2021).

Publications appear under the author name A. Shankar.